WORKED SOLUTION

STEP III 1994, Question 3

Pure • Spheres, Planes and Linking

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q3

The key idea

Each circle lies in its own plane, and the two planes meet in the line L. So the only places where C_1 can pass through the plane of C_2 are its two points on L, and the same is true the other way round. Whether the circles are linked therefore comes down to a one-dimensional question: how the four points on L are arranged along the line.

Solution

Planes and spheres

Let a sphere have centre Q and radius r, and let d be the distance from Q to a plane Π. Let F be the foot of the perpendicular from Q to Π. For a point X of Π, Pythagoras gives $QX^2 = d^2 + FX^2$, so X is on the sphere exactly when $FX^2 = r^2 - d^2$. Hence:

- if $d > r$, the plane misses the sphere;
- if $d = r$, the plane touches the sphere at the single point F (it is a tangent plane);
- if $d < r$, the intersection is a circle in Π with centre F and radius $\sqrt{r^2 - d^2}$. When $d = 0$ this is a great circle, of radius r.

The data

Completing the square, S_2 is

$$x^2 + (y-3)^2 + (z-2)^2 = 3,$$

with centre $(0,3,2)$ and radius $\sqrt{3}$. The sphere S_1 has centre O and radius $\sqrt{7}$.

The distance from O to P_1 is $\frac{|-1|}{\sqrt{10}} = \frac{1}{\sqrt{10}} < \sqrt{7}$, so C_1 is a genuine circle, of radius $\sqrt{7 - \frac{1}{10}}$. The distance from $(0,3,2)$ to P_2 is $\frac{|0-3+1|}{\sqrt{2}} = \sqrt{2} < \sqrt{3}$, so C_2 is a circle of radius $\sqrt{3-2} = 1$. Its centre is the foot of the perpendicular, $(0,3,2) + (1,-1,0) = (1,2,2)$; check that $1 - 2 + 1 = 0$.

The line L

Subtracting the plane equations gives $2x - 2 = 0$, so $x = 1$, and then $y = x + 1 = 2$, while z is free. So

$$L = \{(1, 2, t) : t \in \mathbb{R}\},$$

the vertical line through $(1, 2, 0)$.

On S_1: $1 + 4 + t^2 = 7$, so $t = \pm\sqrt{2}$. Thus L meets S_1 at

$$(1, 2, \sqrt{2}) \quad \text{and} \quad (1, 2, -\sqrt{2}).$$

On S_2: $1 + 1 + (t - 2)^2 = 3$, so $t - 2 = \pm 1$. Thus L meets S_2 at

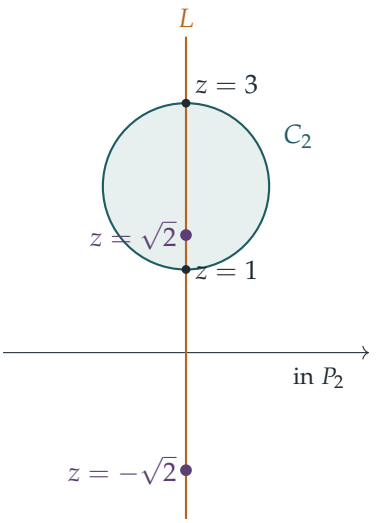
$$(1, 2, 1) \quad \text{and} \quad (1, 2, 3).$$

Since L lies in P_1, the points of L on S_1 are exactly the points of L on C_1; similarly the points of L on S_2 are the points of L on C_2. In particular $(1, 2, 1)$ and $(1, 2, 3)$ are the ends of a diameter of C_2, consistent with its centre $(1, 2, 2)$ and radius 1.

Are the circles linked?

First, the circles do not meet. Any common point lies in $P_1 \cap P_2 = L$, and the points of C_1 on L ($z = \pm\sqrt{2}$) differ from those of C_2 on L ($z = 1, 3$).

Let D_2 be the flat disc in P_2 bounded by C_2. It meets L in the diameter from $(1, 2, 1)$ to $(1, 2, 3)$, that is, in the points $(1, 2, t)$ with $1 \le t \le 3$.



The picture is the plane P_2, with C_2 and its disc shaded. The purple points are where C_1 passes through P_2.

Ask yourself Why does C_1 really pass *through* P_2 at these two points, rather than just touching it?

The line L divides the plane P_1 into two half-planes, and these lie on opposite sides of P_2, because P_1 crosses P_2 along L. The circle C_1 meets L in two distinct points, so L cuts C_1 into two arcs, one in each half-plane. Hence one arc of C_1 lies strictly on one side of P_2 and the other arc strictly on the other side, and C_1 crosses P_2 at $(1, 2, \sqrt{2})$ and at $(1, 2, -\sqrt{2})$, and nowhere else.

Now $1 < \sqrt{2} < 3$, so the crossing at $(1, 2, \sqrt{2})$ is inside the disc D_2. But $-\sqrt{2} < 1$, so the crossing at $(1, 2, -\sqrt{2})$ is outside it. Going once round C_1, we pass through the disc spanned by C_2 exactly once, and come back across P_2 outside the disc. In other words C_1 threads through C_2 once, like two links of a chain.

Along L the four points come in the order

$$-\sqrt{2} \;<\; 1 \;<\; \sqrt{2} \;<\; 3,$$

so the pairs from C_1 and from C_2 *interleave*. That is exactly what makes C_1 cross D_2 once. (If neither pair separated the other, C_1 would cross D_2 either twice or not at all, and the circles would not be linked.) Hence **the circles C_1 and C_2 are linked**.

One cannot pull them apart: any continuous movement taking them to two far-apart circles would have to change the number of times C_1 crosses the disc of C_2 from odd to even. The crossings come and go in pairs unless one circle passes through the other, which a separation without cutting does not allow.

Checks

Substitute each point: for example $(1, 2, 1)$ gives $3 - 2 - 1 = 0$, $1 - 2 + 1 = 0$ and $1 + 4 + 1 - 12 - 4 + 10 = 0$. Numerically, we parametrised both circles and evaluated Gauss's linking integral $\frac{1}{4\pi} \oint\oint \frac{(\mathbf{r}_1 - \mathbf{r}_2) \cdot (\mathrm{d}\mathbf{r}_1 \times \mathrm{d}\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|^3}$, getting 1.002 (an unlinked pair gives 0). We also found that the closest distance between the circles is about 0.09, so they come close but do not touch.

Common slips

- Checking only whether the circles intersect. They don't, but that alone says nothing about linking.
- Forgetting that the crossings of C_1 with P_2 must lie on L, and looking for them elsewhere.
- Getting the centre of S_2 wrong: the $-6y$ and $-4z$ terms give centre $(0, 3, 2)$, not $(0, -3, -2)$.

Beyond STEP

Whether two disjoint closed curves are linked is measured by an integer invariant, the *linking number*. It can be computed by counting signed crossings of one curve through a surface spanning the other, which is what we did, or by Gauss's double integral in the Checks. Linking numbers are a first step into knot theory and algebraic topology. Colin Adams's *The Knot Book* is a very readable introduction.

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